



Rincoin 840k S1 Consensus Change Specification

S1 Block-Subsidy Transition

Rincoin Community Forge

2026-08-07

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1 Status

This is a **Candidate Specification — Discussion Draft** for a Rincoin hard-fork consensus change. Publication does not imply adoption. The specification, reference implementation, tests, activation plan, service readiness, and chain-identification decisions remain under review.

Document version 1.0 · Public-review date 2026-08-07

The key words **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT**, and **MAY** in this document are to be interpreted as described in RFC 2119 and RFC 8174 when, and only when, they appear in all capitals ([Bradner 1997](#); [Leiba 2017](#)).

2 Scope

This specification defines only the maximum permitted block subsidy at and after block height 840,000.

It does not change:

- prior blockchain history or the validity of blocks below height 840,000;
- existing UTXOs or ownership claims;
- RinHash or its parameters;
- Dark Gravity Wave v3;
- transaction format or signature rules;
- transaction-fee accounting;
- any unrelated consensus rule.

Transaction replay, network separation, release deployment, and service migration require separate decisions. They are not solved by this subsidy rule.

3 Definitions

Term	Definition
Activation height H	840000
Reward epoch length L	210000 blocks
Base unit	10^{-8} RIN; 1 RIN = 100000000 base units
Pre-activation reference subsidy R_0	625000000 base units, or 6.25000000 RIN

Term	Definition
Post-activation epoch index k	$\lfloor (h - H)/L \rfloor + 1$ for height $h \geq H$
Maximum permitted block subsidy	Maximum newly issued amount a coinbase MAY claim at a height, excluding transaction fees
Rounding rule	Integer floor after multiplication by 19 and division by 20
First zero-subsidy height	73710000

The epoch index is one-based after activation. Epoch $k = 1$ begins at height 840,000 and ends at height 1,049,999.

4 Pre-Activation Rule

For every block height $h < 840000$, conforming software MUST apply the deployed pre-activation subsidy rule without modification. This specification does not restate that rule as new normative text.

The inspected deployed implementation is `GetBlockSubsidy` at commit `b52c87778f800dc5f4e2f59c372bad` ([Rin-coin contributors 2026](#)). The candidate implementation MUST demonstrate identical subsidy outputs and block-validity behaviour below activation through regression tests.

5 Activation

The candidate rule activates at **block height 840,000**.

- Height 839,999 MUST use the deployed pre-activation rule.
- Height 840,000 MUST use post-activation epoch $k = 1$.
- Height 840,001 MUST remain in post-activation epoch $k = 1$.

There is no transition window in which either subsidy rule is accepted.

6 Post-Activation Subsidy Rule

For block height $h \geq H$, calculate:

$$k = \left\lfloor \frac{h - H}{L} \right\rfloor + 1.$$

Define the epoch subsidy recursively:

$$R_0 = 625,000,000,$$

$$R_i = \left\lfloor \frac{19R_{i-1}}{20} \right\rfloor \quad \text{for } i = 1, 2, \dots, k.$$

The maximum permitted block subsidy at height h MUST be R_k base units.

The recurrence is based on the **previous integer-rounded epoch subsidy**. An implementation MUST NOT independently evaluate a real-number expression such as $R_0(19/20)^k$, because that can produce different integer results.

This is a recursive 19/20 rule, nominally a 5% reduction, with integer-floor rounding after each epoch. The effective percentage reduction can differ slightly from 5% after rounding.

Consequences:

- the first post-activation subsidy is 593750000 base units, or 5.93750000 RIN;
- each epoch subsidy is the integer floor of 19 times the previous integer-rounded subsidy divided by 20;
- there is no fixed subsidy floor;
- there is no constant terminal phase;
- there is no perpetual tail emission;
- integer rounding eventually produces zero;
- once $R_k = 0$, every later R_i remains zero.

7 Integer Arithmetic

A conforming implementation:

1. MUST represent subsidy in integer base units;
2. MUST multiply the previous epoch subsidy by 19 before dividing by 20;
3. MUST use truncating integer division on a non-negative value, which is equivalent to mathematical floor;
4. MUST NOT use floating-point arithmetic in consensus calculation;
5. MUST use an integer type and operation sequence that cannot overflow;
6. MUST produce the same result on every supported architecture and build configuration.

For this schedule, the largest recurrence multiplication is $625000000 * 19 = 11875000000$, which fits safely in a signed 64-bit integer. The reference implementation SHOULD nevertheless use the consensus amount type and explicit checked assumptions rather than relying on platform-sized integer types.

Conceptual pseudocode:

```
CAmount GetPost840kSubsidy(int height)
{
    assert(height >= 840000);

    const int64_t epoch =
        ((int64_t)height - 840000) / 210000 + 1;
```

```

    CAmount subsidy = 625000000;
    for (int64_t i = 0; i < epoch && subsidy > 0; ++i) {
        subsidy = (subsidy * 19) / 20;
    }
    return subsidy;
}

```

The final implementation MAY use an equivalent more efficient method only if it is proven by exhaustive epoch tests to return the same integer value.

8 Issuance Accounting

The maximum permitted subsidy MUST be a deterministic function of block height. It MUST NOT depend on:

- actual subsidy claimed by earlier coinbase transactions;
- circulating, spendable, active, lost, or burned supply;
- UTXO state beyond ordinary block validation;
- market price or network hashrate;
- transaction fees.

The total value of coinbase outputs MUST NOT exceed the maximum permitted block subsidy at height h plus the transaction fees collected from that block.

Transaction fees are transfers of existing RIN and are not newly issued supply. A miner MAY claim less than the maximum permitted subsidy. Any underclaimed amount MUST NOT be reissued at a later height.

9 Boundary Vectors

The following vectors are generated from `data/scenario_config.json` by `scripts/simulate_monetary_scenarios.py`. They cover activation, the first ten post-activation epochs, threshold crossings, the final non-zero epoch, and the first zero-subsidy block.

Vector	Height	Epoch	Recur- rence input	Maximum subsidy, base units	Maximum subsidy, RIN	Boundary purpose
S1-001	839999	legacy	n/a	625000000	6.25000000	last pre- activation block

Vector	Height	Epoch	Recur- rence input	Maximum subsidy, base units	Maximum subsidy, RIN	Boundary purpose
S1-002	840000	1	625000000	593750000	5.93750000	activation block; first block of post- activation epoch 1
S1-003	840001	1	625000000	593750000	5.93750000	second activation- epoch block
S1-004	1049999	1	625000000	593750000	5.93750000	last block of post- activation epoch 1
S1-005	1050000	2	593750000	564062500	5.64062500	first block of post- activation epoch 2
S1-006	1259999	2	593750000	564062500	5.64062500	last block of post- activation epoch 2
S1-007	1260000	3	564062500	535859375	5.35859375	first block of post- activation epoch 3
S1-008	1469999	3	564062500	535859375	5.35859375	last block of post- activation epoch 3
S1-009	1470000	4	535859375	509066406	5.09066406	first block of post- activation epoch 4
S1-010	1679999	4	535859375	509066406	5.09066406	last block of post- activation epoch 4
S1-011	1680000	5	509066406	483613085	4.83613085	first block of post- activation epoch 5

Vector	Height	Epoch	Recur- rence input	Maximum subsidy, base units	Maximum subsidy, RIN	Boundary purpose
S1-012	1889999	5	509066406	483613085	4.83613085	last block of post- activation epoch 5
S1-013	1890000	6	483613085	459432430	4.59432430	first block of post- activation epoch 6
S1-014	2099999	6	483613085	459432430	4.59432430	last block of post- activation epoch 6
S1-015	2100000	7	459432430	436460808	4.36460808	first block of post- activation epoch 7
S1-016	2309999	7	459432430	436460808	4.36460808	last block of post- activation epoch 7
S1-017	2310000	8	436460808	414637767	4.14637767	first block of post- activation epoch 8
S1-018	2519999	8	436460808	414637767	4.14637767	last block of post- activation epoch 8
S1-019	2520000	9	414637767	393905878	3.93905878	first block of post- activation epoch 9
S1-020	2729999	9	414637767	393905878	3.93905878	last block of post- activation epoch 9
S1-021	2730000	10	393905878	374210584	3.74210584	first block of post- activation epoch 10
S1-022	2939999	10	393905878	374210584	3.74210584	last block of post- activation epoch 10

Vector	Height	Epoch	Recur- rence input	Maximum subsidy, base units	Maximum subsidy, RIN	Boundary purpose
S1-023	8189999	35	109265378	103802109	1.03802109	last block before crossing below 1 RIN
S1-024	8190000	36	103802109	98612003	0.98612003	crossing below 1 RIN
S1-025	8190001	36	103802109	98612003	0.98612003	second block after crossing below 1 RIN
S1-026	11129999	49	53285986	50621686	0.50621686	last block before crossing below 0.5 RIN
S1-027	11130000	50	50621686	48090601	0.48090601	crossing below 0.5 RIN
S1-028	11130001	50	50621686	48090601	0.48090601	second block after crossing below 0.5 RIN
S1-029	17639999	80	10865369	10322100	0.10322100	last block before crossing below 0.1 RIN
S1-030	17640000	81	10322100	9805995	0.09805995	crossing below 0.1 RIN
S1-031	17640001	81	10322100	9805995	0.09805995	second block after crossing below 0.1 RIN

Vector	Height	Epoch	Recur- rence input	Maximum subsidy, base units	Maximum subsidy, RIN	Boundary purpose
S1-032	73500000	347	2	1	0.00000001	first one- base-unit subsidy block
S1-033	73709999	347	2	1	0.00000001	final non-zero subsidy block
S1-034	73710000	348	1	0	0.00000000	first zero- subsidy block
S1-035	73710001	348	1	0	0.00000000	second zero- subsidy block

Machine-readable forms:

- `data/S1_normative_test_vectors.csv`
- `data/S1_normative_test_vectors.json`

Artifact	SHA-256
<code>data/S1_normative_test_vectors.csv</code>	750eb0396661f0284ff2d3837093b4f5f97556b15937d950f08de2501
<code>data/S1_normative_test_vectors.json</code>	54d11a4b4b1551cbe1b7eaf21346bdb14e25b2612af97c8339ba376

A conforming reference implementation **MUST** match every vector. Before activation readiness is claimed, an independent implementation **SHOULD** generate the same outputs without importing the simulator’s subsidy function.

10 Maximum Scheduled Issuance

Under the candidate integer rule:

- maximum scheduled issuance below height 840,000 is 19,687,500.00000000 RIN;
- maximum scheduled issuance at and above height 840,000 is 24,937,493.01750000 RIN;
- total maximum scheduled issuance is **44,624,993.01750000 RIN**;
- the first zero-subsidy height is **73,710,000**.

The original whitepaper’s 168,000,000 RIN statement is a historical upper design bound, not the effective maximum of this rule. No state-dependent mechanism can increase scheduled issuance from 44,624,993.01750000 RIN toward 168,000,000 RIN.

These totals assume every block claims the maximum permitted subsidy. Actual issued and circulating supply MAY be lower.

11 Compatibility and Activation Risk

At height 840,000, the candidate rule permits 5.93750000 RIN, while the unchanged deployed binary-halving rule permits 3.12500000 RIN. A block whose coinbase claims more than the lower amount but no more than the candidate amount will be valid under the candidate rule and invalid under unchanged software.

Nodes enforcing different rules can therefore diverge at or after the first block whose coinbase claim distinguishes the rules. Deployment **MUST** be treated as a hard fork. Implementers **MUST** test:

- height 839,999 and 840,000 boundary behaviour;
- coinbase claims below, equal to, and one base unit above each maximum;
- reorganisation across the activation boundary in both directions;
- initial block download across the boundary;
- block-template subsidy and miner coinbase validation;
- deterministic behaviour on all supported platforms.

12 Chain Identification and Replay

This subsidy change separates block validity but does not, by itself, prevent transaction replay between branches that share transaction formats and signature rules. A block-version marker alone would not provide transaction replay protection.

Before activation readiness is claimed, a separate deployment analysis **MUST** resolve or explicitly accept:

- P2P and software identity;
- wallet and service chain selection;
- explorer and exchange labelling;
- deposit and withdrawal routing;
- transaction replay exposure;
- whether replay protection requires an additional consensus change.

The chain-identification and replay-risk mechanism (a scheduled coinbase commitment plus a sighash-level `sig_fork_id`) is specified in `technology/consensus-transition.md` §5–6 and `technology/replay-protection-plan.md`; it applies identically across S1, S5/b, and S6/b and is not repeated in this document.

13 Reference Implementation

The candidate implementation is for testing and public review. It is not final, adopted, production-ready, official, or activation-ready.

The public implementation home is the `rincoin-community` GitHub organization. A testing-mode-only candidate implementation is published at `rincoin-community/rincoin-core`, branch `consensus/s1-testing`, commit `73eb5b3ce` (CI run, concluded green). This branch refuses to start on mainnet unless `RINCOIN_TESTING_ALLOW_MAINNET=1` is set in the process environment at every invocation; it is not for production use.

Build instructions: `./scripts/prereqs.sh && ./scripts/build.sh` (see that repository's `README.md` for platform-specific detail).

Test-vector cross-check, executed test results, and the cross-implementation comparison against older Rincoin releases and other independently-developed forks: `verification/S1_Consensus_Testing_Summary.qmd` and its evidence annex, `verification/S1/`. Local verification is complete and passing (532 C++ unit tests, 6 fork-scenario functional tests, 159 pre-existing functional tests, all green); the full cross-implementation producer $\times$ validator matrix and two narrow commitment test-vector cases remain outstanding — see that annex's own `open-items.md`. The `branch_id` used in the tests matches `technology/consensus-transition.md` §5's canonical synthetic value.

14 Non-Normative Rationale

The companion *Rincoin Monetary Policy and Security Scenario Analysis* compares the candidate against the unchanged rule, extended epochs, fixed phases, cap-derived decline, Customized Halving study schedules, and perpetual tails. The *Rincoin Monetary Review Summary* presents the decision path and questions for public review ([Rincoin Community Forge 2026a](#), [2026b](#)).

S1 allocates the remaining issuance through recursive 19/20 reductions, nominally 5%. It is one member of the principal comparison pair in the public-review package; S5/b allocates almost the same final issuance through approximately four-year binary integer halvings, nominally 50%. This rationale is non-normative and does not alter any requirement in this specification.

References

- Bradner, Scott. 1997. *Key Words for Use in RFCs to Indicate Requirement Levels*. RFC 2119. Internet Engineering Task Force. <https://doi.org/10.17487/RFC2119>.
- Leiba, Barry. 2017. *Ambiguity of Uppercase Vs Lowercase in RFC 2119 Key Words*. RFC 8174. Internet Engineering Task Force. <https://doi.org/10.17487/RFC8174>.

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